# Does Domain-Specific Embedding Fine-Tuning Improve RAG Performance on Financial Document QA? A Multi-Model Study

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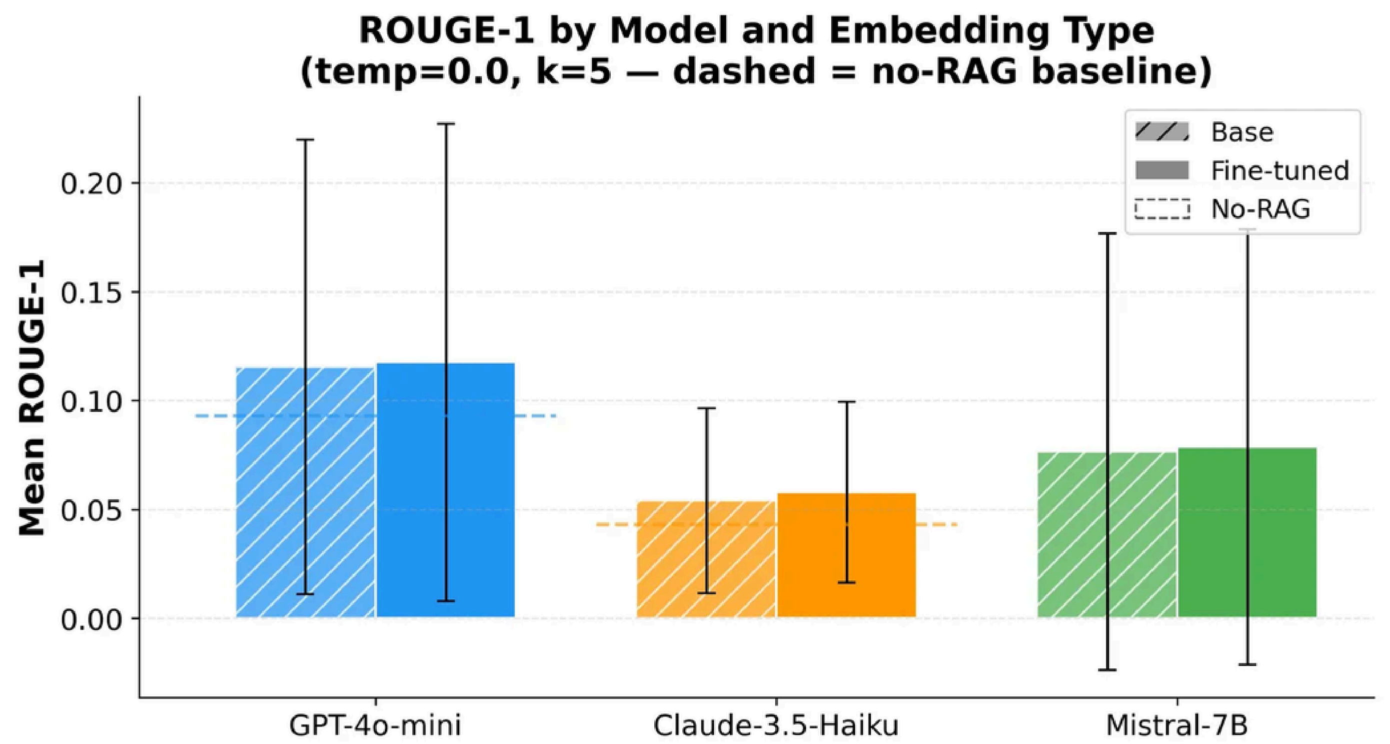
## ABSTRACT

We tested whether fine-tuning retrieval embeddings improves LLM performance on financial document QA. Retrieval quality drives most of the gains, embedding tier matters far less than expected, and numerical accuracy remains an open problem.

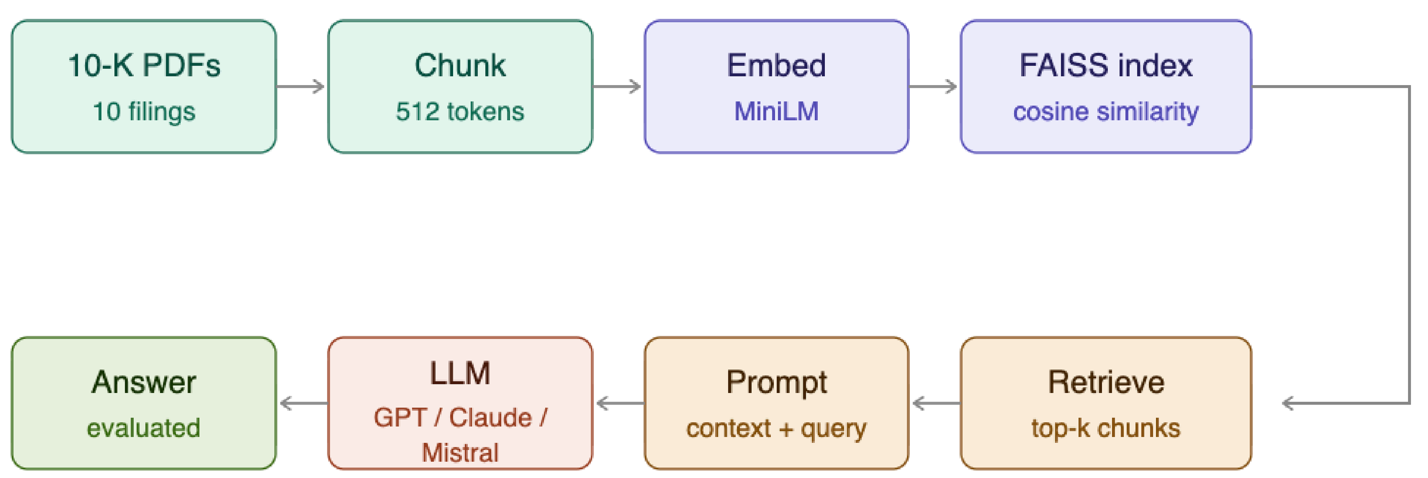



## INTRODUCTION

- 10-K filings are dense, table-heavy, and full of domain-specific language that challenges even capable models. RAG grounds answers in retrieved document content, but the embedding model driving retrieval is often overlooked.
- Does fine-tuning embeddings on financial text improve downstream QA quality?
- Does the effect vary across model capability tiers?

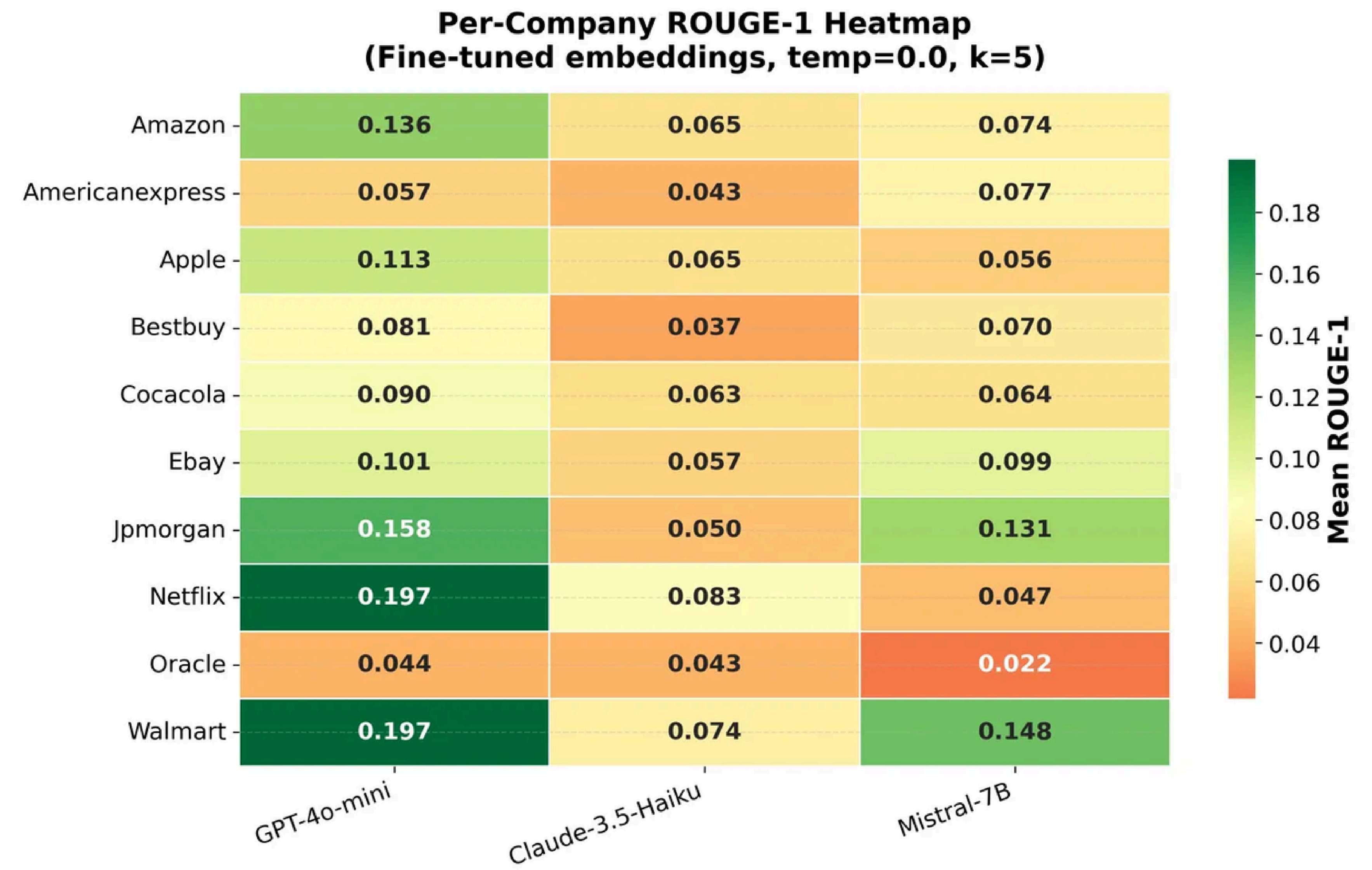
## METHODS

- 150 questions across 10 companies' 2022 10-K filings, factual, numerical, and reasoning. Fine-tuned MiniLM-L6-v2 on 35 financial question-passage pairs (3 epochs).
- Models: GPT-4o-mini, Claude-3.5-Haiku, Mistral-7B (4-bit quantized, Colab T4)
- Varied embedding type, temperature (0.0 / 0.7), and retrieval depth (k = 3 / 5)
- Evaluated using ROUGE-1, numeric accuracy (±10% tolerance), and LLM Judge, a 0–3 quality score assigned by GPT-4o-mini as an automated evaluator




## RESULTS

- Adding retrieval is the single largest improvement in the pipeline. Claude's LLM Judge score jumps +0.834 from RAG alone
- Claude leads on LLM Judge (2.067) but ranks last on ROUGE-1, ROUGE penalizes verbose but correct answers
- The bottleneck is retrieval quality, not embedding tier




| Model | Embedding | ROUGE-1 | ROUGE-L | LLM Judge (0–3) | Numeric Acc. | Latency (s) |
|---|---|---|---|---|---|---|
| GPT-4o-mini | Base | 0.115 | 0.107 | 1.933 | 21.6% | 2.20 |
| GPT-4o-mini | Fine-tuned | 0.118 | 0.108 | 1.940 | 13.5% | 2.78 |
| Claude-3.5-Haiku | Base | 0.054 | 0.048 | 2.067 | 18.9% | 2.92 |
| Claude-3.5-Haiku | Fine-tuned | 0.058 | 0.053 | 2.067 | 16.2% | 2.90 |
| Mistral-7B | Base | 0.076 | 0.070 | 1.247 | 13.5% | 4.78 |
| Mistral-7B | Fine-tuned | 0.079 | 0.073 | 1.307 | 5.4% | 4.85 |
| GPT-4o-mini | No-RAG | 0.093 | – | 1.387 | – | – |
| Claude-3.5-Haiku | No-RAG | 0.043 | – | 1.233 | – | – |

Table 1: Main results (temp=0.0, *k*=5). Exact Match was 0.0 for all conditions (omitted). Numeric Acc. = numeric accuracy on numerical questions (±10% tolerance). Mistral latency reflects local GPU inference; GPT-4o-mini and Claude-3.5-Haiku reflect API round-trip time.

## FAILURE ANALYSIS

- Mistral fails at 3x the rate of Claude and 2.5x the rate of GPT. Three failure modes account for most errors:
- Retrieval miss, answer absent from retrieved chunks; model correctly abstains but scores 0
- Wrong number retrieved, correct section, wrong table row; primary driver of numeric accuracy drop
- Context overflow, Mistral only. preventable with k=3

| Model | Fact. | Num. | Tot. | Primary Failure |
|---|---|---|---|---|
| GPT-4o-mini | 10 | 10 | 20 | Retrieval miss; wrong line item |
| Claude-3.5-Haiku | 6 | 8 | 14 | Verbose non-answers |
| Mistral-7B | 34 | 23 | 57 | Context overflow; wrong number |

## CONCLUSION

- Retrieval quality is the dominant factor in this pipeline, fine-tuning the embedding model is secondary. Numerical QA over financial documents remains largely unsolved even under optimal conditions.
- ROUGE alone is misleading; always pair it with a semantic metric
- Future work: table, aware chunking and structured number extraction